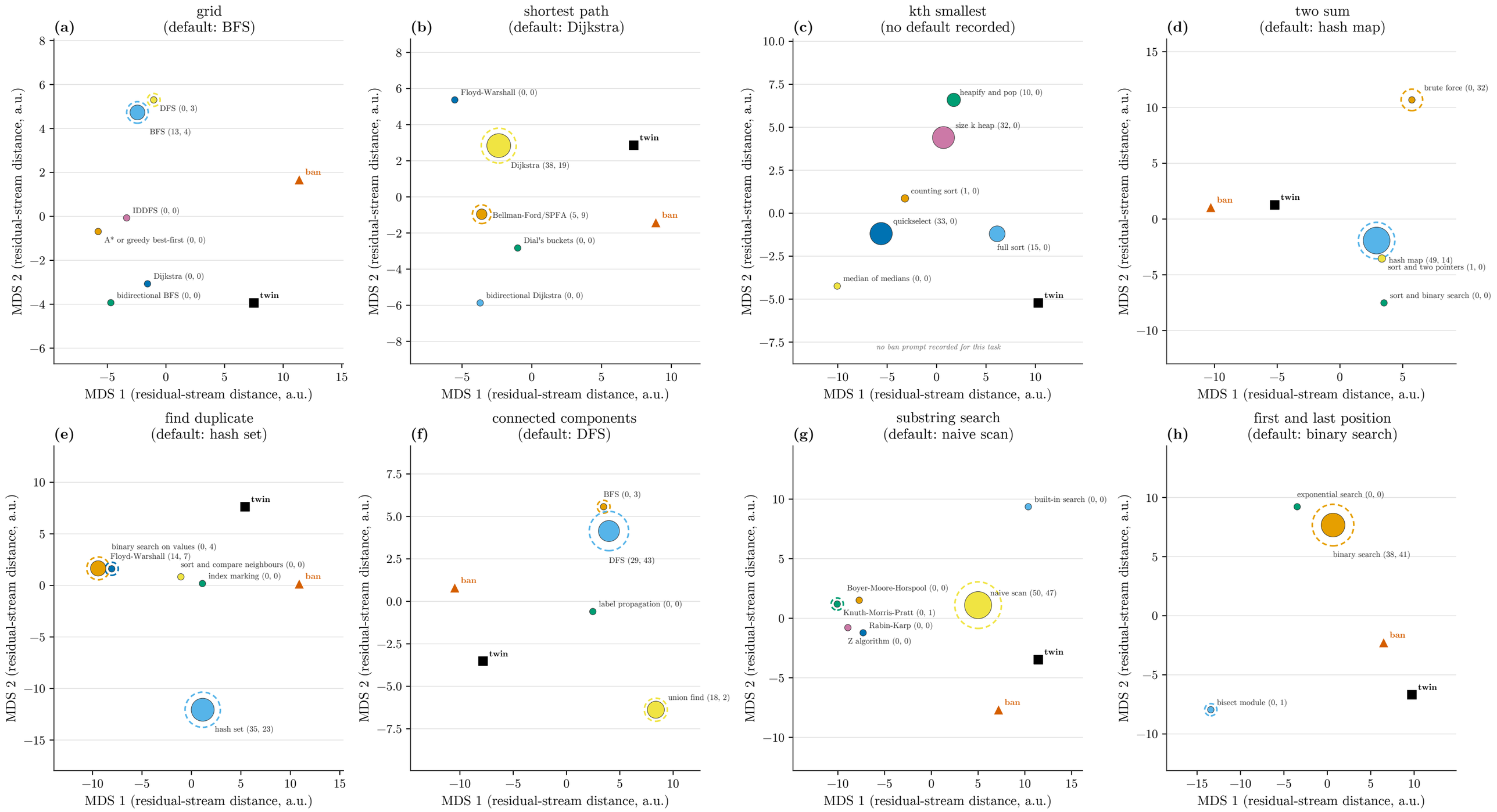
(a)
grid
(default: BFS)
DFS (0, 3)
BFS (13, 4)
ban
IDDFS (0, 0)
A* or greedy best-first (0, 0)
Dijkstra (0, 0)
bidirectional BFS (0, 0)
twin
(b)
shortest path
(default: Dijkstra)
Floyd-Warshall (0, 0)
twin
Dijkstra (38, 19)
Bellman-Ford/SPFA (5, 9)
ban
Dial's buckets (0, 0)
bidirectional Dijkstra (0, 0)
(c)
kth smallest
(no default recorded)
heapify and pop (10, 0)
size k heap (32, 0)
counting sort (1, 0)
quickselect (33, 0)
full sort (15, 0)
median of medians (0, 0)
twin
no ban prompt recorded for this task
(d)
two sum
(default: hash map)
brute force (0, 32)
ban
twin
hash map (49, 14)
sort and two pointers (1, 0)
sort and binary search (0, 0)
(e)
find duplicate
(default: hash set)
twin
binary search on values (0, 4)
Floyd-Warshall (14, 7)
sort and compare neighbours (0, 0)
index marking (0, 0)
ban
hash set (35, 23)
(f)
connected components
(default: DFS)
BFS (0, 3)
DFS (29, 43)
ban
label propagation (0, 0)
twin
union find (18, 2)
(g)
substring search
(default: naive scan)
built-in search (0, 0)
Boyer-Moore-Horspool (0, 0)
naive scan (50, 47)
Knuth-Morris-Pratt (0, 1)
Rabin-Karp (0, 0)
Z algorithm (0, 0)
twin
ban
(h)
first and last position
(default: binary search)
exponential search (0, 0)
binary search (38, 41)
ban
twin
bisect module (0, 1)
MDS 1 (residual-stream distance, a.u.)
MDS 2 (residual-stream distance, a.u.)
algorithm centroid (small = few twin programs)
algorithm centroid (large = many twin programs)
dashed ring = programs under the ban
twin centroid
ban centroid
label "name (twin, ban)" = programs implementing it per condition